

Standardization, Simplification,
Specialisation and Interchangeability

STANDARDISATION

Standards are at the base of all mass production. They make possible thousands of different articles to be placed within the reach of everybody.

- *Standards* convey the sense that there are only certain specific sizes made and sold.
- Standards are carefully established specifications for products, materials, etc.
- *Standardization* means producing maximum variety of products from the minimum variety of (i.e., *standardized*) materials, parts, tools and processes.
- Standardization is the process of establishing standards or units of measure by which extent, quality, quantity, value, performance, etc., may be compared and measured.

STANDARDISATION

- It is the setting up of standards for quantity, quality, raw material, sizes and performance etc. of any product. It is done after considering various scientific procedures and is helpful for checking the quality performance and value of product
- Standard of product can only be achieved by standardizing tools equipments and materials etc
- All the tools, equipment, working conditions, material, quantity etc. should be standardized.
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STANDARDISATION PROCEDURE

- (a) With the help of market research, sales statistics, etc. decide what to sell in future.
 - (b) Then, define a standard range of products.
 - (c) From the range, ask the designer to develop minimum variety of components to match the range.
- Introduce new materials, components, etc. if necessary.
- An approach to standardization necessitates the classification of materials and component parts.

CLASSIFICATION

- A system of *classification and coding* is necessary for the design of new products within the range defined.

Such a system should readily :

- (i) Identify and locate identical items.
 - (ii) Facilitate the use of standard items in new designs.
 - (iii) Identify substitutes in case of *stock outs*.
 - (iv) Help developing Group Technology.
 - (v) Aid to improve parts location in the store.
- *Classification procedure* involves the following steps :
 - (i) Define all items.
 - (ii) Classify each item according to its basic characteristics.
 - (iii) Identify each item by allocating to it some meaningful code number.

Advantages of Stndardisation

- 1. Interchangeability of products is possible.
- 2. Better quality of products.
- 3. Use of automatic machines is easily and economically possible.
- 4. It is possible to provide guarantee period for the performance of product.
- 5. It saves largely production time and labour and thus cost of products is reduced.
- 6. It helps in reduction of stock investment and storages space.
- 7. It helps in reduction of maintenance spares and thereby reduces maintenance cost.
- 8. It provides improved managerial control and better manufacturing efficiency.
- 9. Better customer's satisfaction is possible.

Disadvantages

- Reduction in choice because of reduced variety and consequent loss of business or custom.
- Changes in public taste seriously affect a company producing only standardized product range.
- It becomes difficult to introduce new models because of less flexible (existing) production facilities and due to the high cost of specialised production equipment.
- Standardization tends to favour large famous companies, because small or new concerns can rarely get much business even by producing same items and by selling them at the same price as the big companies.
- Standards once set, resist change and thus standardization may become an obstacle to progress.

SIMPLIFICATION

- Simplification is the process of reducing the variety of products manufactured (known as variety reduction).
- Simplification is concerned with the reduction of product range, assemblies, parts, materials and design.
- A manufacturer may reduce the number of different types of radio sets from a dozen to three or four to simplify his range.
- Simplification makes a product, assembly or design, simpler, less complex or less difficult.
- Simplification removes the superfluous. It decreases variety of sizes ; for example a garment-factory making tea-shirts in sizes 16, $16\frac{1}{4}$, $16\frac{1}{2}$, $16\frac{3}{4}$, 17, $17\frac{1}{4}$ etc., can eliminate superfluous sizes such as $16\frac{1}{4}$, $16\frac{3}{4}$, $17\frac{1}{4}$, etc., and thus simplify its production line.
- A production line is generally simplified when it possesses unnecessary complexity and confusion.

SPECIALISATION

- Specialization is the natural outcome of the application of standardization and simplification.
- Specialization means concentrating efforts on a particular field of action or towards a specific attempt.
- A worker is said to be *specialized* in a work when he acquires skill and proficiency in it by concentrating solely on it (*i.e.*, on that particular work or job). A mechanic, brick-layer or an engineer is a specialist in his field.
- A factory producing spark plugs only is a specialist in its production.
- Specialization as applied to human activities on shop floor can be defined as 'Division of Labour'. This means that if a worker instead of completing the full product, performs one small operation on the product and attains proficiency in that one activity, he becomes a specialist in that.

Specialisation

- Advantages

- (1) Workers achieve a high state of skill and proficiency.
- (2) They take smaller times to complete the activity in which they are specialized.
- (3) Thus they raise their salaries and their standard of living.

- Disadvantages

- (1) Specialized labour and equipment are not flexible, *i.e.*, they cannot be used for other purposes.
- (2) Specialization may result in monotony.

INTERCHANGEABILITY

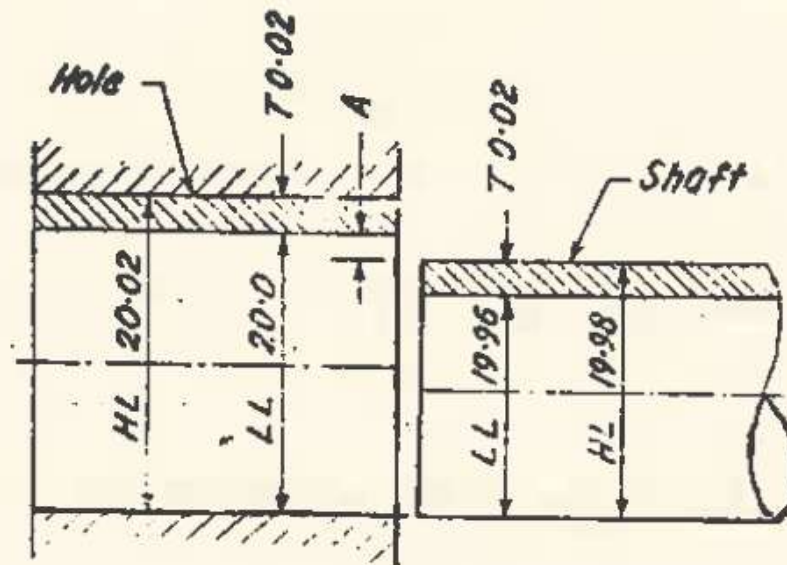
- *Interchangeability* or *interchangeable manufacture* means that any standardized component will assemble correctly with any mating component, both being chosen at random.

For an *interchangeable system* to work, the parts produced should be as near identical as possible, and for continuous production, a transfer line will achieve this best because it eliminates the human control of the machines.

- Interchangeability reduces cost because the task of assembly is simplified. Moreover standard replacement parts can be drawn from the stock with the certainty that they will fit without alteration.
- In order to achieve interchangeability :
 - (i) Appropriate component tolerances must be specified (from the standard) to suit the type of fit required.
 - (ii) Manufacturing process should be selected to make components within the specified tolerances.
 - (iii) A system of inspection and quality control should check that only components within the specified tolerances are accepted for use.

Interchangeability Continued

- In interchangeable system is also called a *limit system* or system of *limits and fits*.
Fig. 5.3, gives the concept of limits, tolerance and allowance.
- The larger and smaller dimensions of the hole or shaft are called the **LIMITS**, there is a *high limit* (HL) and a *low limit* (LL).



HL : High limit
 LL : Low limit
 T : Tolerance
 A : Allowance

Fig. 5.3. Limits and tolerances for hole and shaft.

- The difference between the high and low limits (which is the margin allowed for variations in workmanship) is known as **TOLERANCE** (T).